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INST 414

Final Report

# People Only Care When It Happens: Attention and Cybersecurity News Cycle

## *Introduction*

In September 2017, Equifax announced that hackers had accessed the personal information of approximately 147 million Americans, including Social Security numbers, birth dates, and credit card information. The breach dominated headlines. Within weeks, the public had moved on. This study investigates whether that pattern holds across major cyberattacks, or whether high-profile incidents ever produce lasting shifts in public concern. I hypothesize they do not. Rather than generating long-term upward trends in public attention, I expect to find short, sharp peaks that fade as quickly as they appear, because if the data confirms what the Equifax case suggests, the uncomfortable conclusion is that when it comes to digital security, most people simply do not stay concerned long enough to demand change.

## *Data and Methods*

To examine this question, I used Google Trends data from 2017 to 2024, tracking US search volume for terms like “ransomware,” “cybersecurity,” and “data breach” across web and news searches. I focused on three incidents: the Equifax data breach (September 2017), the SolarWinds hack (December 2020), and the Colonial Pipeline ransomware attack (May 2021).

These represent different cyber threats: consumer data theft, government espionage, and critical infrastructure attack, spread across the dataset. Time-series analysis, K-Means clustering, and linear regression evaluated changes in search behavior before and after each event.

## *Results*

Figure 1 shows the time-series results, confirming the hypothesis. Each incident causes a spike in web search interest, then declines to pre-incident levels within weeks. The September 2017 Equifax breach peaked sharply and subsided by October. The May 2021 Colonial Pipeline attack caused a similar spike. The SolarWinds hack, however, prompted less public response despite affecting federal agencies, likely because it unfolded gradually without a dramatic disclosure, which usually drives search behavior. This difference is significant and will be discussed further.

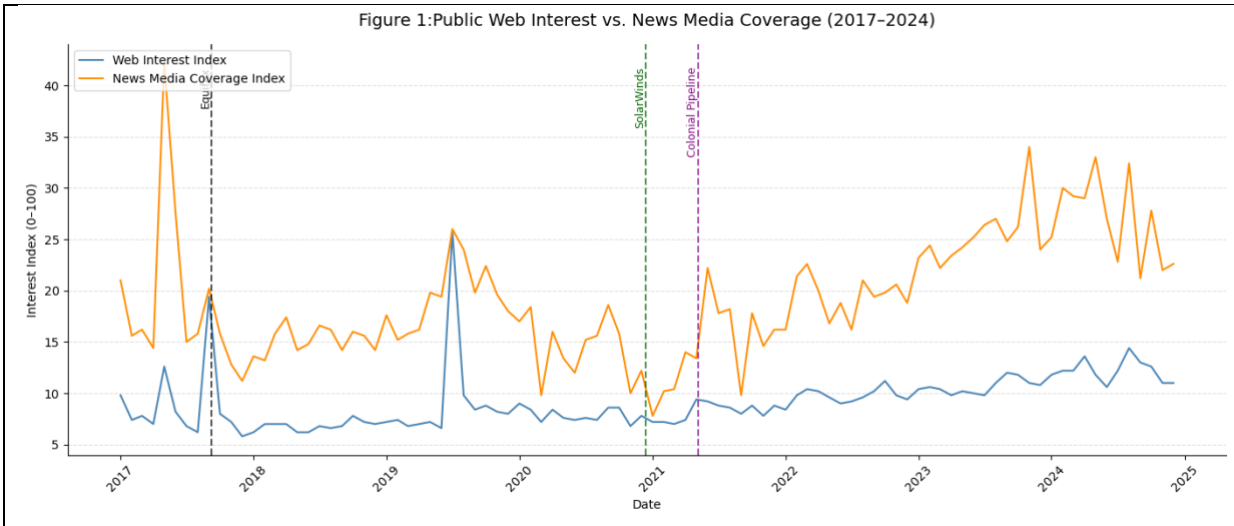


*Figure 1. Web search interest spikes around each major incident then returns to baseline within weeks. Despite rising news media coverage since 2022, public search interest has remained flat, suggesting that increased reporting alone is no longer enough to sustain public engagement.*

Also in Figure 1, a trend emerges after 2022: news media coverage of cybersecurity topics has increased steadily, while public web search interest has remained largely flat. This divergence suggests that the relationship between media coverage and public engagement is weakening over time, a finding examined more directly in Figure 3.

To understand how public attention shifts, I used K-Means clustering to group weeks by interest and media coverage. Figure 2 shows three states: a large cluster of weeks with minimal engagement; a second with elevated concern; and a tiny cluster, just two points out of nearly 400 weeks, indicating peak panic. This supports the time-series results. During eight years of major cyberattacks, public concern peaked for about two weeks. For companies and policymakers, this narrow window of accountability is crucial to avoid prolonged scrutiny.

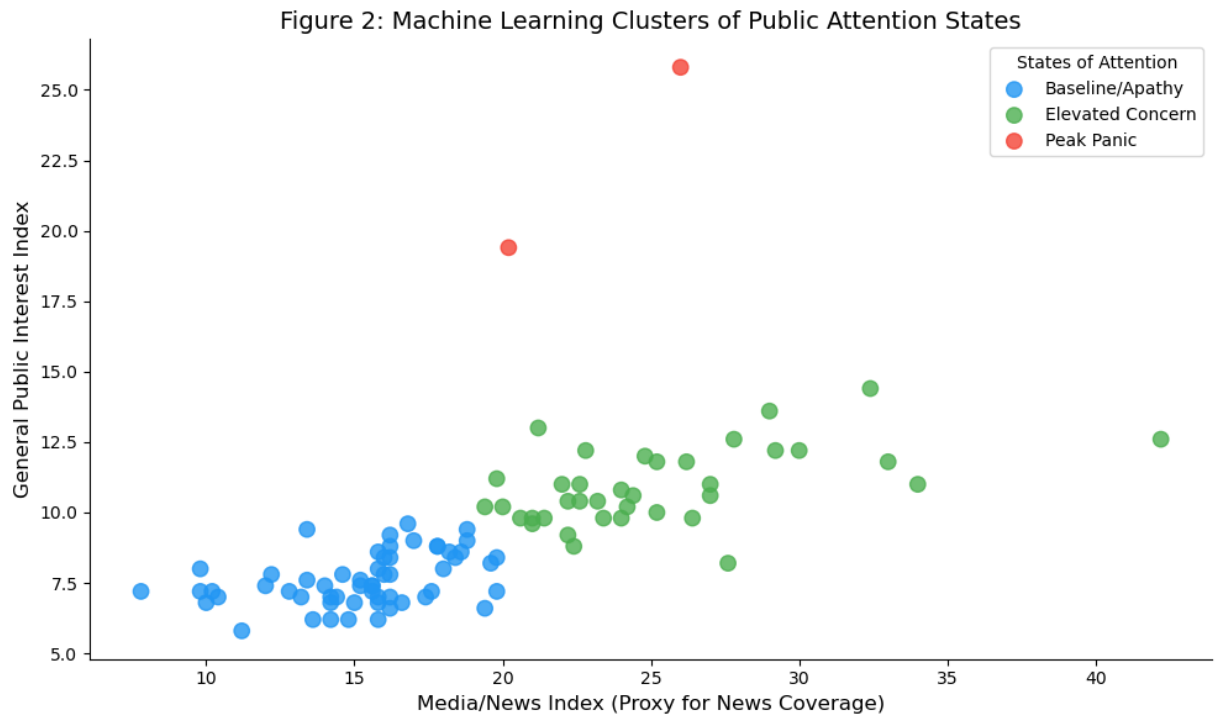


*Figure 2. K-Means clustering identifies three states of public attention across the 2017-2024 period: Baseline/Apathy, Elevated Concern, and Peak Panic. Only two weeks out of nearly 400 qualify as Peak Panic, highlighting how rarely cyber incidents drive public concern to its maximum level.*

Figure 3 more directly examines the relationship between news media coverage and public search interest. The results indicate a statistically significant positive relationship between the two variables; weeks with higher media coverage of cybersecurity topics are associated with higher levels of public search interest. This finding aligns with the expectation that media coverage drives public attention. However, as noted above, this relationship appears to be weakening since 2022. Media coverage of cybersecurity is more frequent than at any prior point in the dataset, yet public interest has not increased. This suggests that coverage alone is no longer sufficient to generate sustained engagement and that the public may be experiencing saturation with cybersecurity-related news.

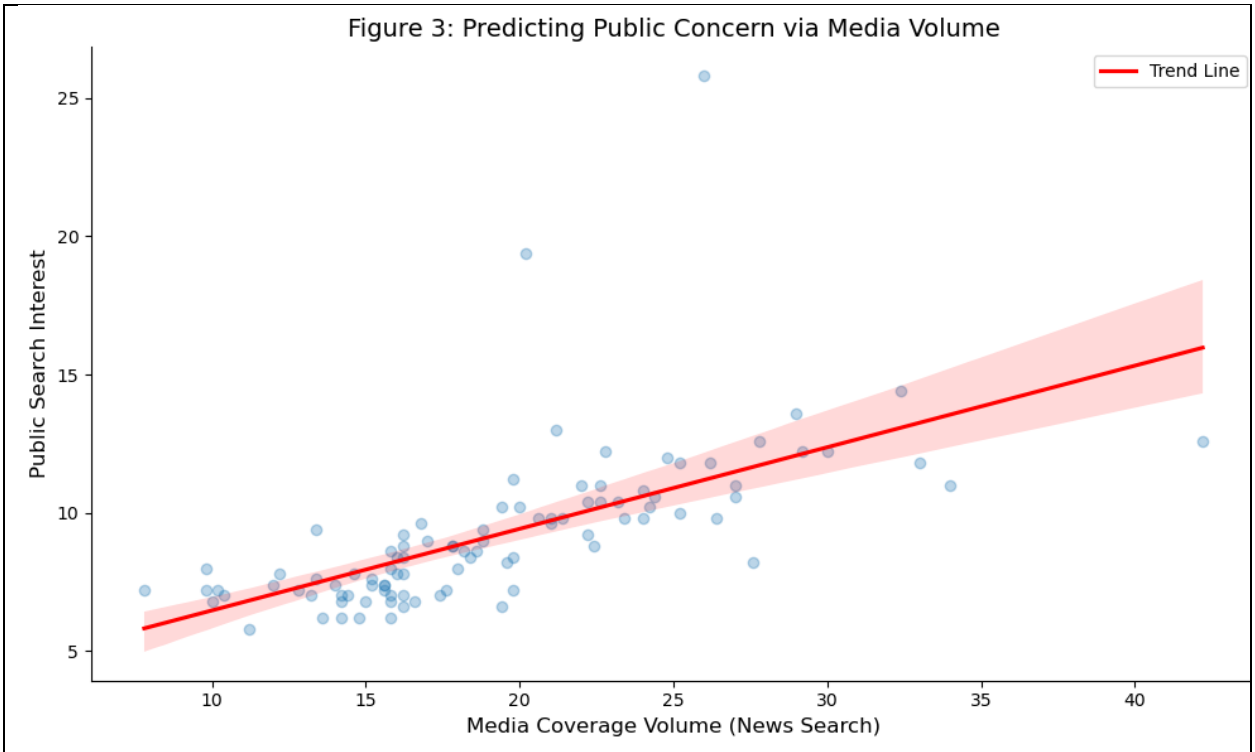


*Figure 3. News media coverage is a statistically significant predictor of public search interest, though this relationship has weakened since 2022, as rising coverage has failed to produce a corresponding rise in public engagement.*

Figure 4 quantifies the rate of attention decay after each incident. For both Equifax and Colonial Pipeline, public interest declined measurably within 30 days of its peak. For SolarWinds, a 60-day window was used to account for the slower buildup of public awareness associated with that incident; even within this extended window, the same pattern of decline is observed. In each case, public interest returned to a level comparable to or lower than pre-incident baselines within a relatively short period. The consistency of this pattern across three incidents of different types and severities is notable. It suggests that the rapid decay of public attention following cyber events is not a function of any particular incident's characteristics but reflects a more general tendency in how the public responds to cybersecurity news.

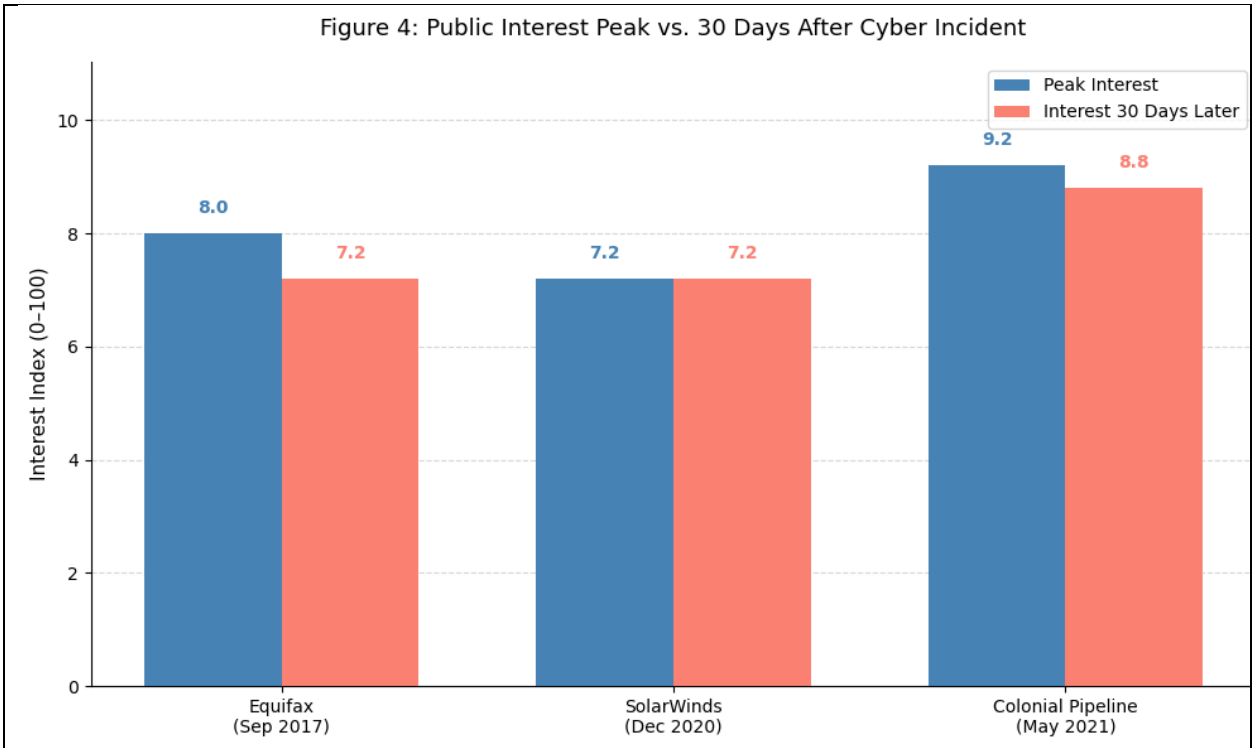


*Figure 4. Public search interest declined within 30 days of peak for all three incidents, suggesting that attention decay after cyberattacks is a consistent pattern rather than a case-by-case occurrence.*

*Discussion*

Taken together, these findings support the hypothesis that public attention to cybersecurity is episodic rather than cumulative. Major incidents generate short-term spikes in concern that dissipate quickly, leaving no lasting increase in baseline engagement. This pattern has direct implications for how cybersecurity accountability is structured. If public pressure is the primary mechanism by which corporations are held responsible for data security failures and by which policymakers are motivated to enact reform, then a public that disengages within weeks of a major incident is insufficient to sustain that accountability. The problem is compounded by the SolarWinds case, which suggests that the most sophisticated and potentially damaging attacks may be precisely those least likely to generate the kind of dramatic response that drives search behavior.

*Conclusion*

This study does not argue that the public is indifferent to cybersecurity. Spikes in interest after each incident prove that people pay attention when something goes wrong. The problem is that attention fades before anything changes. This is not a new phenomenon; the 2014 Sony hack generated the same cycle of outrage and amnesia, meaning we have been watching this pattern repeat for over a decade and still have not figured out how to break it.

At some point, the conversation has to shift from "why won't the public stay engaged" to "why are we still depending on the public to do the work that regulators should be doing?" Mandatory breach-disclosure timelines, automatic regulatory reviews following significant incidents, and minimum security standards for critical infrastructure should not require a news

cycle to trigger them. Companies should not get to decide how seriously they take cybersecurity based on whether anyone is currently paying attention. The whole point of regulation is that it functions even when the public moves on, which, as this study confirms, happens faster than most people realize.

The data is clear. Public attention spikes, fades, and resets. The threats do not. If accountability keeps waiting for the public to care, it will arrive too late.